Enhanced Disease Search Capabilities

We are implementing a two-pronged strategy to significantly improve disease search accuracy:1. Improved Synonym Matching

- **Richer Embeddings:** We will enhance the text used to generate AI embeddings by incorporating comprehensive data, including disease names, all known synonyms, and detailed definitions. This provides a much richer context for the model, dramatically improving its ability to accurately match related terms and synonyms.

2. Hybrid Search (Exact + Semantic)

The new search process combines speed with intelligence through a two-step approach:

- **Step 1: Prioritized Exact Match:** The system will first check an in-memory index for an instant, direct match between the user's query and a known disease name or synonym. If an exact match is found, it is prioritized instantly.
- **Step 2: Intelligent Semantic Search:** A vector-based semantic search is then performed to find other contextually related diseases.

This hybrid model delivers the highest accuracy and performance, offering both the speed and perfect precision of a direct lookup *and* the intelligent, context-aware results of a semantic search.

Steps performed while execution :-

Faster Startup via Pre-computation and Caching

- **Initial Setup:** The first time the server runs, it generates vector embeddings for over 14,000 diseases. These embeddings are saved to `backend/data/disease_embeddings.npy`. This one-time process is expected to be time-consuming.
- **Subsequent Startups:** All future server startups will be extremely fast (taking only seconds) as the embeddings are loaded directly from the saved file.

Optimized Embedding Generation

- **Parallel Processing:** The initial, one-time embedding generation is now significantly faster due to the implementation of parallel processing.
- **Richer Context:** To improve semantic matching and provide better suggestions, the content used to generate the embeddings has been expanded. It now includes the disease name, all its synonyms, and its definition, providing the model with more context.

Hybrid Search for Superior Accuracy

The search functionality now employs a hybrid approach:

1. **Exact Match:** The system first searches for exact matches in disease names and synonyms. For instance, searching for "trichomonas infection" will instantly return "Trichomonas Infections" with a perfect match score.
2. **Semantic Search (Fallback/Augmentation):** If no exact matches are found, or to offer additional suggestions, the system then performs a semantic search using vector embeddings to find diseases that are conceptually similar.

Comparison of Retrieval Methods: Current Semantic Search vs. RAG

Our Current Semantic Search

- **Mechanism:** Queries are converted into vector embeddings (using a model via Ollama). This query embedding is compared for similarity (cosine similarity) against precomputed embeddings of disease records.
- **Role of LLM:** The LLM is used *only* for creating the embeddings (vectorization), not for generating the final answer.
- **Output:** Returns a structured list of top disease matches, including the name, ID, and similarity score.
- **Result Format:** Structured data only; no free-text answer.

Retrieval-Augmented Generation (RAG)

- **Mechanism:** A user query is vectorized and used to retrieve relevant documents or passages (similar to your semantic search). This retrieved content is then supplied as context to a generative LLM (e.g., GPT, Llama).
- **Role of LLM:** The LLM uses the retrieved content to synthesize a custom, free-text answer. It can quote, summarize, or explain the information.
- **Output:** A natural language, context-rich answer.
- **Result Format:** Free-text, capable of combining, summarizing, or explaining the retrieved data, often with citations.

Key Differences Summarized

Feature	Your Semantic Search	RAG (Retrieval-Augmented Generation)
Retrieval	Yes (Vector search)	Yes (Vector or Keyword search)
Generation	No	Yes (LLM generates the answer)
Output Type	Structured data (Disease list)	Free-text, context-rich answer
LLM for Answer?	No (Used only for embeddings)	Yes (Uses retrieved documents as context)
Example Use	"Top 5 most similar diseases"	"Based on the data, Disease X is related to..."